

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé V – English Source-Aligned Cumulative Assembly Checkpoint

Fourteen independently sealed bounded units; French lines 1770–2136

19 July 2026

Evidence boundary. This is a coherent reflowed assembly of the mathematical bodies of fourteen independently source-reviewed and sealed working units. The corrected French arXiv TeX is the translation authority; the body occupies physical source-PDF pp. 55–63 / running pp. 47–55 and was checked in the bounded reviews. Physical pp. 52–54 / running pp. 44–46 are retained only as predecessor context. The current jcreinhold e7a259f English Markdown is one comparison lineage only. This package passed independent cumulative audit on 19 July 2026. It remains a bounded checkpoint, not a critical edition, source certification, or release.

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Exposé V. Local Duality and Structure of the $H^i(M)$

1. Complexes of homomorphisms

1.1

Let $F^\bullet$ and $G^\bullet$ be two graded modules. Write

$$\mathrm{Hom}^\bullet(F^\bullet, G^\bullet) \quad (1)$$

for the graded module of homomorphisms of graded modules from $F^\bullet$ to $G^\bullet$. Thus

$$\mathrm{Hom}^s(F^\bullet, G^\bullet) = \prod_k \mathrm{Hom}(F^k, G^{k+s}). \quad (2)$$

Suppose that $F^\bullet$ (respectively $G^\bullet$) is a complex, and let d_1 (respectively d_2) be its differential. For $h \in \mathrm{Hom}^s(F^\bullet, G^\bullet)$, set¹

$$d(h) = h \circ d_1 + (-1)^{s+1} d_2 \circ h. \quad (3)$$

One checks immediately that $d \circ d = 0$; hence $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$, equipped with d , is a complex. The cohomology group of this complex is denoted by

$$\mathbb{H}^\bullet(F^\bullet, G^\bullet). \quad (4)$$

If $G^\bullet$ is injective in every degree, then

$$F^\bullet \mapsto \mathbb{H}^\bullet(F^\bullet, G^\bullet)$$

is an exact ∂ -functor. Likewise, for arbitrary $F^\bullet$,

$$G^\bullet \mapsto \mathbb{H}^\bullet(F^\bullet, G^\bullet)$$

is an exact δ -functor on the category of complexes $G^\bullet$ that are injective in every degree.

Remark 1.2. *The cycles of $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ that commute or anticommute with the differentials, according to the degree. The boundaries of $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ that are homotopic to zero.*

¹*Editor's note.* The original sign convention was different, but it is incompatible with the convention of Exposé VIII. The latter seems more reasonable because in that case the degree-zero cohomology is the set of homotopy classes of morphisms from $F^\bullet$ to $G^\bullet$. The calculations below have been modified accordingly.

Let A be a ring, let M (respectively N) be an A -module, and let $R(M)$ (respectively $R(N)$) be an injective resolution of M (respectively N). There is then a canonical isomorphism²

$$\mathbb{H}^s(R(M), R(N)) \simeq \text{Ext}^s(M, N). \quad (1.3)$$

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Indeed, let $i: M \rightarrow R(M)$ be the canonical augmentation, and let $h \in \text{Hom}^s(R(M), R(N))$. Denote by t_s the map

$$h \longmapsto h_0 \circ i$$

from $\text{Hom}^s(R(M), R(N))$ to $\text{Hom}(M, R(N)^s)$. The family $(t_s)_{s \geq 0}$ defines a homomorphism of (ordinary) complexes³

$$t: \text{Hom}^\bullet(R(M), R(N)) \longrightarrow \text{Hom}^\bullet(M, R(N));$$

that is,

$$(dh)_0 \circ i = d_2 \circ h_0 \circ i.$$

One checks easily that, on passing to cohomology, t induces an isomorphism. It follows in particular that

$$\mathbb{H}^\bullet(R(M), R(N))$$

does not “depend” on the chosen injective resolution $R(M)$ (respectively $R(N)$) of M (respectively N).

To every exact sequence of A -modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \quad (5)$$

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one associates an exact sequence of injective resolutions

$$0 \longrightarrow R(M') \longrightarrow R(M) \longrightarrow R(M'') \longrightarrow 0. \quad (6)$$

One checks that the isomorphism (1.3) commutes with the homomorphisms

$$\mathbb{H}^s(R(M'), R(N)) \longrightarrow \mathbb{H}^{s+1}(R(M''), R(N)), \quad (8)$$

$$\text{Ext}^s(R(M'), R(N)) \longrightarrow \text{Ext}^{s+1}(R(M''), R(N)), \quad (9)$$

deduced from (6) and (5).⁴

Let P be a third A -module, and let $R(P)$ be an injective resolution of P . Then composition of graded morphisms gives a pairing

$$\text{Hom}^i(R(N), R(M)) \times \text{Hom}^j(R(M), R(P)) \longrightarrow \text{Hom}^{i+j}(R(N), R(P)), \quad (10)$$

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which defines a pairing

$$\mathbb{H}^i(R(N), R(M)) \times \mathbb{H}^j(R(M), R(P)) \longrightarrow \mathbb{H}^{i+j}(R(N), R(P)), \quad (11)$$

and hence a homomorphism of functors in M :

$$\mathbb{H}^i(R(N), R(M)) \longrightarrow \text{Hom}\left(\mathbb{H}^j(R(M), R(P)), \mathbb{H}^{i+j}(R(N), R(P))\right). \quad (1.4)$$

We shall see that (1.4) is a homomorphism of δ -functors in M . The exact sequences (5) and (6) give a commutative diagram:

²*Editor's note.* The strange original numbering has been preserved.

³Here M also denotes the complex $M[0]$ consisting of M placed in degree zero.

⁴*Source note.* The corrected French TeX and the direct compiled source page both use resolution arguments in (9): $R(M')$ and $R(N)$ in the source term, and $R(M'')$ and $R(N)$ in the target term. This bounded translation reproduces that literal reading; it does not silently replace those arguments by M' , M'' , and N .

$$\begin{array}{ccc}
\mathrm{Hom}^i(R(N), R(M')) & \longrightarrow & \mathrm{Hom}\left(\mathrm{Hom}^j(R(M'), R(P)), \mathrm{Hom}^{i+j}(R(N), R(P))\right) \\
\downarrow & & \downarrow \mathrm{Hom}(q, \mathrm{id}) \\
\mathrm{Hom}^i(R(N), R(M)) & \longrightarrow & \mathrm{Hom}\left(\mathrm{Hom}^j(R(M), R(P)), \mathrm{Hom}^{i+j}(R(N), R(P))\right) \\
\downarrow p & & \downarrow \\
\mathrm{Hom}^i(R(N), R(M'')) & \longrightarrow & \mathrm{Hom}\left(\mathrm{Hom}^j(R(M''), R(P)), \mathrm{Hom}^{i+j}(R(N), R(P))\right).
\end{array}$$

Let $h \in \mathrm{Hom}^i(R(N), R(M''))$ (respectively $g \in \mathrm{Hom}^j(R(M'), R(P))$) be a cycle, and let $h' \in \mathrm{Hom}^i(R(N), R(M))$ (respectively $g' \in \mathrm{Hom}^j(R(M), R(P))$) be such that $p(h') = h$ (respectively $q(g') = g$). To say that (1.4) is a homomorphism of δ -functors in M is to say that

$$g \circ dh' - dg' \circ h \quad (12)$$

is a coboundary in $\mathrm{Hom}^\bullet(R(N), R(P))$.

Now

$$\begin{aligned}
dh' &= h' \circ d_1 + (-1)^{i+1} d_2 \circ h', \\
dg' &= g' \circ d_2 + (-1)^{j+1} d_3 \circ g',
\end{aligned}$$

with the obvious notation. Thus (12) becomes

$$g \circ h' \circ d_1 + (-1)^{i+1} g \circ d_2 \circ h' - g' \circ d_2 \circ h - (-1)^{j+1} d_3 \circ g' \circ h.$$

On the other hand, since h and g are cycles,

$$\begin{aligned}
g \circ d_2 &= (-1)^j d_3 \circ g, \\
d_2 \circ h &= (-1)^i h \circ d_1.
\end{aligned}$$

Consequently (12) is finally written as

$$d(g \circ h' + (-1)^{i+1} g' \circ h),$$

which completes the proof.

Thus (1.3) and (1.4) give a homomorphism of δ -functors in M :

$$\mathrm{Ext}^i(N, M) \longrightarrow \mathrm{Hom}\left(\mathrm{Ext}^j(M, P), \mathrm{Ext}^{i+j}(N, P)\right). \quad (1.5)$$

2. The local duality theorem for a regular local ring

Let A be a regular local ring of dimension r , let $\mathfrak{m}$ be the maximal ideal of A , and let M be a finitely generated A -module. Set

$$H^i(M) = H_{\mathfrak{m}}^i(M) \quad (\text{so } H^i(M) = \varinjlim_k \mathrm{Ext}^i(A/\mathfrak{m}^k, M)).$$

We saw in IV, 5.4 that $I = H^r(A)$ is a dualizing module for A ; denote the associated dualizing functor by D . In (1.5), put $N = A/\mathfrak{m}^k$ and $P = A$. We then obtain a homomorphism of δ -functors in M ,

$$\varphi_k: \mathrm{Ext}^i(A/\mathfrak{m}^k, M) \longrightarrow \mathrm{Hom}\left(\mathrm{Ext}^{r-i}(M, A), \mathrm{Ext}^r(A/\mathfrak{m}^k, A)\right). \quad (13)$$

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Passing to the direct limit over k gives a homomorphism of δ -functors

$$\varphi: H^i(M) \longrightarrow D\left(\operatorname{Ext}^{r-i}(M, A)\right). \quad (14)$$

Theorem 2.1 (Local duality theorem). The preceding functorial homomorphism φ is an isomorphism.

Proof. If $i > r$, the right-hand side of (14) is trivially zero, and the left-hand side is zero because

$$H^i(M) = \varinjlim_k \operatorname{Ext}^i(A/\mathfrak{m}^k, M),$$

and each $\operatorname{Ext}^i(A/\mathfrak{m}^k, M)$ is zero (syzygy theorem).

If $i = r$, the preceding discussion shows that the two functors in M , $H^r(M)$ and $D(\operatorname{Hom}(M, A))$, are right exact. Since A is noetherian and M is finitely generated, it is enough to verify the isomorphism for $M = A$, where it is immediate.

To show that φ is a functorial isomorphism, it now suffices to proceed by descending induction on i and observe that every finitely generated module admits a finite presentation, while for $i < r$ both sides of (14) vanish when M is finitely generated and free. This is clear for the right-hand side. Since H^i commutes with finite direct sums, for the left-hand side it is enough to show that $H^i(A) = 0$ for $i < r$. This follows from III, 3.4, since $\operatorname{prof}(A) = r$. $\square$

3. Application to the structure of the $H^i(M)$

Theorem 3.1. Let A be a Noetherian local ring, let D be a dualizing functor for A , and let $M \neq 0$ be a finitely generated A -module of dimension n . Then:

- (i) $H^i(M) = 0$ if $i < 0$ or if $i > n$.
- (ii) $D(\widehat{H^i(M)})$ is a finitely generated $\widehat{A}$ -module of dimension at most i .
- (iii) $H^n(M) \neq 0$; moreover, if A is complete, then $D(H^n(M))$ has dimension n and

$$\operatorname{Ass}(D(H^n(M))) = \{ \mathfrak{p} \in \operatorname{Ass}(M) \mid \dim(A/\mathfrak{p}) = n \}.$$

Proof. Let I be the dualizing module associated with D . We know that $\widehat{I}$ is a dualizing module for $\widehat{A}$. On the other hand,

$$\begin{aligned} \widehat{H^i(M)} &= H^i(\widehat{M}), \\ D(\widehat{H^i(M)}) &= \operatorname{Hom}(H^i(\widehat{M}), \widehat{I}), \\ \dim \widehat{M} &= \dim M. \end{aligned}$$

Consequently, we may suppose that A is complete. By Cohen's theorem, every complete local ring is a quotient of a regular local ring. To reduce to that case, we need the following lemma.

Lemma 3.2. Let X (respectively Y) be a ringed space, let X' (respectively Y') be a closed subspace of X (respectively Y), and let $f: X \rightarrow Y$ be a morphism of ringed spaces such that $f^{-1}(Y') = X'$. Let F be an $\mathcal{O}_X$ -module. Denote by A (respectively B) the ring $\Gamma(\mathcal{O}_X)$ (respectively $\Gamma(\mathcal{O}_{Y'})$), and by $\bar{f}: B \rightarrow A$ the ring homomorphism corresponding to f . There is a spectral sequence of B -modules with initial term

$$E_2^{p,q} = H_{Y'}^p(Y, R^q f_*(F)), \quad (15)$$

converging to the B -module $H_{X'}^\bullet(X, F)_{[\bar{f}]}$.

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Proof. Let $\mathcal{O}_{Y,Y'}$ be the sheaf $\mathcal{O}_Y|_{Y'}$, extended by zero outside Y' (see Exposé I). There is an isomorphism of B -modules

$$\mathrm{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)) \simeq \mathrm{Hom}(f^*(\mathcal{O}_{Y,Y'}), F)_{[\bar{f}]}.$$
 (16)

Now

$$f^*(\mathcal{O}_{Y,Y'}) = \mathcal{O}_{X,X'}.$$
 (17)

Moreover, if G is an injective $\mathcal{O}_X$ -module, then $f_*(G)$ is an injective $\mathcal{O}_Y$ -module, at least when f is flat; one can easily reduce to this case by replacing $\mathcal{O}_X$, and so forth, by the constant sheaves of rings $\mathbb{Z}$. Hence the spectral sequence of the composite functor

$$F \longmapsto \mathrm{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)),$$

whose initial term is

$$E_2^{p,q} = \mathrm{Ext}^p(Y; \mathcal{O}_{Y,Y'}, R^q f_*(F)),$$

converges, in view of (16) and (17), to

$$\mathrm{Ext}^\bullet(X; \mathcal{O}_{X,X'}, F)_{[\bar{f}]}.$$

The lemma then follows from Exposé I, Proposition 2.3 bis.

Q.E.D.

Now let $f: B \rightarrow A$ be a surjective homomorphism of local rings. Let

$$f: \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(B)$$

be the corresponding morphism of affine schemes. Set $X = \mathrm{Spec}(A)$ (respectively $X' = \{\mathfrak{m}_A\}$) and $Y = \mathrm{Spec}(B)$ (respectively $Y' = \{\mathfrak{m}_B\}$), and let M be an A -module and $\widetilde{M}$ the corresponding $\mathcal{O}_X$ -module. Since $R^q f_*(\widetilde{M}) = 0$ for $q > 0$, the spectral sequence (15) degenerates, and Lemma 3.2 gives an isomorphism of B -modules

$$H_{\{\mathfrak{m}_B\}}^n(Y, f_*(\widetilde{M})) \simeq H_{\{\mathfrak{m}_A\}}^n(X, \widetilde{M})_{[f]},$$
 (18)

hence an isomorphism of B -modules

$$H_{\mathfrak{m}_B}^n(M_{[f]}) \simeq H_{\mathfrak{m}_A}^n(M)_{[f]}.$$
 (19)

On the other hand, if D_A (respectively D_B) is the dualizing functor for A (respectively B), then

$$D_A(M)_{[f]} \simeq D_B(M_{[f]}).$$
 (20)

Finally, since there is an isomorphism of rings

$$B / \mathrm{Ann} M_{[f]} \simeq A / \mathrm{Ann} M,$$
 (21)

it follows that the change of base rings under consideration changes nothing. We may therefore suppose that A is regular of dimension r .

By (2.1), we have

$$D(H^i(M)) = \mathrm{Ext}^{r-i}(M, A).$$
 (22)

We shall prove the equivalence of the following properties:

- (a) $\dim \mathrm{Ext}^j(M, A) \leq r - j$;
- (b) for every $\mathfrak{p} \in X = \mathrm{Spec}(A)$ such that $\dim A_{\mathfrak{p}} < j$, one has $\mathrm{Ext}^j(M, A)_{\mathfrak{p}} = 0$;
- (c) $\mathrm{codim}(\mathrm{Supp}(\mathrm{Ext}^j(M, A)), X) \geq j$.

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To prove (a) $\implies$ (b), let $\mathfrak{p} \in X$ with $\dim A_{\mathfrak{p}} < j$. Then $\dim A/\mathfrak{p} > r - j$, and hence, by (a),

$$\text{Ann}(\text{Ext}^j(M, A)) \not\subset \mathfrak{p},$$

which implies that $\text{Ext}^j(M, A)_{\mathfrak{p}} = 0$. Let $\mathfrak{p} \in \text{Supp}(\text{Ext}^j(M, A))$. Then $\text{Ext}^j(M, A)_{\mathfrak{p}} \neq 0$, so by (b) we have $\dim A_{\mathfrak{p}} \geq j$. Therefore

$$\text{codim}(\text{Supp}(\text{Ext}^j(M, A)), X) = \inf\{\dim A_{\mathfrak{p}} \mid \mathfrak{p} \in \text{Supp}(\text{Ext}^j(M, A))\} \geq j,$$

that is, (b) $\implies$ (c). Finally, (c) implies (a) trivially.

Let us now prove the theorem.

- (i) Let $x = (x_1, \dots, x_r)$ be a system of parameters for A such that $x_i \in \text{Ann } M$ for $i = 1, \dots, r - n$. Let $K^\bullet((x^k), M)$ be the Koszul complex. It is easy to see that the map

$$K^i((x^k), M) \longrightarrow K^i((x^{k'}), M)$$

for $k < k'$ is zero if $i > n$. It follows that

$$H^i(M) = \varinjlim H^i((x^k), M) = 0$$

if $i > n$. On the other hand, it is trivial that $H^i(M) = 0$ if $i < 0$; hence (i) is proved.

- (ii) Since A is regular, $\dim A_{\mathfrak{p}} < j$ implies that the global homological dimension of $A_{\mathfrak{p}}$ is strictly less than j , and hence

$$\text{Ext}^j(M, A)_{\mathfrak{p}} = \text{Ext}_{A_{\mathfrak{p}}}^j(M_{\mathfrak{p}}, A_{\mathfrak{p}}) = 0.$$

Thus we have proved (b), and consequently (a). Assertion (ii) then follows from (22) and (a).

- (iii) There exists a $\mathfrak{p} \in \text{Supp}(M)$ such that $\dim A_{\mathfrak{p}} = r - n$ and such that $\text{Supp}(M_{\mathfrak{p}}) = \{\mathfrak{m}_{A_{\mathfrak{p}}}\}$. Since $A_{\mathfrak{p}}$ is regular if A is, we have $\text{depth } A_{\mathfrak{p}} = r - n$, and therefore

$$\text{Ext}_A^{r-n}(M, A)_{\mathfrak{p}} = \text{Ext}_{A_{\mathfrak{p}}}^{r-n}(M_{\mathfrak{p}}, A_{\mathfrak{p}}) \neq 0. \quad (23)$$

Taking (22) into account, this implies, on the one hand,

$$H^n(M) \neq 0,$$

and, on the other hand,

$$\dim D(H^n(M)) \geq n,$$

hence, by (ii),

$$\dim D(H^n(M)) = n.$$

Now let $Y = \text{Supp}(M)$. By (i), we know that $D(H^n(M')) = \text{Ext}^{r-n}(M', A)$ is a left exact functor in M' on the category $(\mathcal{C}_Y)^\circ$. Hence there exists an A -module H and an isomorphism of functors in M' :

$$\text{Ext}^{r-n}(M', A) = \text{Hom}(M', H).$$

Let Y_i , $i = 1, \dots, k$, be the irreducible components of Y of maximum dimension. We shall see that the assertion $\text{Ext}^{r-n}(M', A) \neq 0$ is equivalent to the assertion that there exists an i such that $\text{Supp } M' \supset Y_i$. Indeed, if $\text{Supp } M' \supset Y_i$, then $\dim(M') = n$, and hence $\text{Ext}^{r-n}(M', A) \neq 0$.

If $\text{Supp } M' \not\supset Y_i$ for every $i = 1, \dots, k$, then $\dim M' < n$, and

$$D(H^n(M')) = \text{Ext}^{r-n}(M', A) = 0.$$

Since $\text{Ass}(\text{Ext}^{r-n}(M, A)) = \text{Supp } M \cap \text{Ass}(H)$, we see that the last assertion of (iii) follows from the following lemma:

Lemma 3.3. *Let $X = \text{Spec}(A)$, let Y be a closed subset of X , let*

$$T: (\mathcal{C}_Y)^\circ \longrightarrow \mathbf{Ab}$$

be a left exact functor, and let Y_i , $i = 1, \dots, k$, be a family of irreducible components of Y such that the assertion $T(M) = 0$ is equivalent to the assertion that, for every i , $\text{Supp } M \not\supset Y_i$. Then T is representable by a module H such that

$$\text{Ass}(H) = \bigcup_{i=1}^k \{y_i\},$$

where y_i is the generic point of Y_i , for $i = 1, \dots, k$.

Proof. For $y \in Y$, construct an A -module $M(y)$ such that

$$\text{Supp}(M(y)) = \overline{\{y\}}.$$

Suppose that $y \neq y_i$ for every $i = 1, \dots, k$. Then $Y_i \not\subset \text{Supp}(M(y))$ for every $i = 1, \dots, k$, and hence $T(M(y)) = 0$. It follows that

$$\text{Ass}(T(M(y))) = \text{Supp}(M(y)) \cap \text{Ass}(H) = \emptyset,$$

and therefore $y \notin \text{Ass}(H)$. If $y = y_i$, then $Y_i \subset \text{Supp}(M(y))$, so $T(M(y)) \neq 0$, and hence

$$\text{Ass}(T(M(y))) = \text{Supp}(M(y)) \cap \text{Ass}(H) \neq \emptyset.$$

By the first part of the proof, this implies $y \in \text{Ass}(H)$, which proves the lemma. Q.E.D.

Example 3.4. *Let A be a noetherian ring, let $X = \text{Spec}(A)$, and let Y be a closed subset of X such that $X - Y$ is affine. Then, for every irreducible component Y_α of Y , one has*

$$\text{codim}(Y_\alpha, X) \leq 1.$$

Proof. Consider X as a prescheme over X . Let $y_\alpha \in Y_\alpha$ be a generic point, and consider the morphism

$$\text{Spec}(\mathcal{O}_{X, y_\alpha}) \longrightarrow X.$$

The affine scheme obtained by base-changing X to $\text{Spec}(\mathcal{O}_{X, y_\alpha})$ is canonically isomorphic to $\text{Spec}(\mathcal{O}_{X, y_\alpha})$.

By (EGA I 3.2.7), if y_0 is the unique closed point of

$$Y_0 = \text{Spec}(\mathcal{O}_{X, y_\alpha}),$$

then $Y_0 - y_0$ is affine. By (EGA III 1.3.1), one obtains

$$H^i(Y_0 - y_0, \mathcal{O}_{Y_0}) = 0 \quad \text{if } i > 0.$$

Hence, by (I 2.9),

$$H^{i-1}(\mathcal{O}_{X, y_\alpha}) = H^i_{\{y_0\}}(Y_0, \mathcal{O}_{Y_0}) = 0 \quad \text{if } i \geq 2.$$

By 3.1(iii), it follows that

$$\dim \mathcal{O}_{X, y_\alpha} \leq 1,$$

and therefore

$$\text{codim}(Y_\alpha, X) = \inf_{y \in Y_\alpha} \dim \mathcal{O}_{X, y} \leq 1.$$

Q.E.D.

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Let A be a noetherian local ring, let $\mathfrak{m}$ be its maximal ideal, and let M be a finitely generated A -module. Suppose that A is a quotient of a regular local ring. Set $X = \operatorname{Spec}(A)$ and, for every $x \in X$, set $\mathfrak{m}_x = \mathfrak{m}A_x$.

Proposition 3.5. *The following two conditions are equivalent:*

- (a) $H^i(M)$ has finite length;
- (b) for every $x \in X - \{\mathfrak{m}\}$,

$$H_{\mathfrak{m}_x}^{i - \dim \overline{\{x\}}}(M_x) = 0.$$

Proof. In view of (3.2), we may suppose that A is regular. By (2.1), we have

$$H^i(M) = D(\operatorname{Ext}^{r-i}(M, A)),$$

where $r = \dim A$. By (IV 4.7), condition (a) is equivalent⁵ to

$$\operatorname{Ext}^{r-i}(M, A) \text{ has finite length.} \quad (24)$$

Condition (24) is equivalent to

$$\text{for every } x \in X - \{\mathfrak{m}\}, \quad \operatorname{Ext}_A^{r-i}(M, A)_x = 0. \quad (25)$$

On the other hand, A_x is regular of dimension $r - \dim \overline{\{x\}}$. Hence, by (2.1),

$$\begin{aligned} H_{\mathfrak{m}_x}^{i - \dim \overline{\{x\}}}(M_x) &= D\left(\operatorname{Ext}_{A_x}^{(r - \dim \overline{\{x\}}) - (i - \dim \overline{\{x\}})}(M_x, A_x)\right) \\ &= D\left(\operatorname{Ext}_{A_x}^{r-i}(M_x, A_x)\right). \end{aligned} \quad (26)$$

Since M is finitely generated, one has

$$\operatorname{Ext}_A^{r-i}(M, A)_x = \operatorname{Ext}_{A_x}^{r-i}(M_x, A_x),$$

which proves the proposition.

Corollary 3.6. *For $H^i(M)$ to have finite length for $i \leq n$, it is necessary and sufficient that*

$$\operatorname{depth}(M_x) > n - \dim \overline{\{x\}}$$

for every $x \in X - \{\mathfrak{m}\}$.

Proof. This follows from (3.5) and (III 3.1).

⁵Indeed, one must show that, if E is a finitely generated A -module and $D(E)$ has finite length, then E has finite length. Let K (respectively Q) be the kernel (respectively cokernel) of the canonical morphism $\epsilon: E \rightarrow DD(E)$. The canonical morphism $\gamma: D(E) \rightarrow DDD(E)$ satisfies $D(\epsilon) \circ \gamma = \operatorname{id}_{D(E)}$. Since $D(E)$ has finite length, γ is an isomorphism, and hence so is $D(\epsilon)$. Since D is exact, it follows that $D(K)$ and $D(Q)$ are zero. It is enough to prove that an A -module whose dual is zero is itself zero, for then $E = DD(E)$ has finite length, just as $D(E)$ does. Indeed, let M_0 be a finitely generated submodule of such a module M . Since D is exact, $D(M_0)$ is a quotient of $D(M)$ and is therefore zero. Again by exactness, $D(M_0/\mathfrak{m}_A M_0) = 0$; by biduality, the finite-length module $M_0/\mathfrak{m}_A M_0$ is therefore zero. Nakayama's lemma then gives $M_0 = 0$, and finally $M = 0$.